

Hassan AL KHAWALDEH

*Mechanical Engineering & Mechanics
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Curriculum Vitae

Education

- 2021–Ongoing **Ph.D. in Mechanical Engineering**, *Lehigh University*, Bethlehem, PA, USA.
Dissertation title: *Physics-Based and Data-Driven Predictive Modeling and Control of Plasma Density in Tokamaks*.
Advisor: Dr. Eugenio Schuster.
- 2017–2021 **B.S. in Mechanical Engineering**, *Lehigh University*, Bethlehem, PA, USA.
Minors: Energy Engineering and Business Management.

Awards

- 2021 **Admitted into Lehigh President’s Scholar Program, which offers students with academic excellence one year of tuition-free study.**
- 2017 **Received the King’s Scholar Scholarship, which is a full-ride award covering the total cost of attendance for a four-year undergraduate program.**

Research Interest

- Topics Adaptive Control (model-reference, self-tuning); Optimal Control (LQR, MPC); Robust Control (H_∞ , structured uncertainty); Nonlinear Control (feedback linearization, Lyapunov-based methods); System Identification (Koopman operator theory, DMD, subspace methods, gray-box modeling); Reinforcement Learning (Model-free, safe RL); Neural-Network Modeling (physics-informed, black-box).

Academic and Industry Engagements

- 2020–Ongoing **Research Assistant**, Plasma Control Group, Lehigh University, Bethlehem, PA
- Engineered large-scale data pipelines from 537 high-fidelity simulations across 41 experiments — totaling 286,477 time slices with 1,147 variables per slice — to train deep neural network surrogate models for real-time control systems.
 - Designed and trained 5 deep neural network surrogate models in Python to approximate high-fidelity physics calculations, achieving $\sim 10^2$ – $10^4\times$ faster predictions ($R^2 = 0.95$, $RMSE = 0.22$) — enabling real-time deployment in compute-constrained environments.
 - Constructed physics-informed models of plasma dynamics — combining first-principles and data-driven models to predict system behavior — enabling $\sim 10^2$ – $10^3\times$ faster control-architecture testing and a sim-to-real transfer pipeline.
 - Designed and benchmarked 4 multi-actuator control architectures (MPC, adaptive, RL-based, LQI) with Kalman-based estimation to identify deployment-ready solutions for accessing and sustaining high-performance plasma scenarios under safety constraints.
 - Integrated and deployed adaptive and safe RL controllers into 2 live experimental control systems alongside engineering and physics teams at General Atomics (GA), Korea Institute of Fusion Energy (KFE), and Princeton Plasma Physics Laboratory (PPPL).
 - Mentored 3 junior researchers on surrogate-model and control-development workflows, covering large-dataset preparation, deep neural network training, and simulation integration — all delivered findings at peer-reviewed conferences.

- Jan–Dec 2020 **Consultant, Capstone: Dorman Industries**, Lehigh University, Bethlehem, PA
- Led a multidisciplinary student team to design an improved diesel engine exhaust gas recirculation (EGR) cooler using CAD tools and heat transfer principles.
 - Developed and iterated detailed CAD models for a redesigned EGR cooler to address thermal and structural performance limitations.
 - Conducted finite element (FEA) and computational fluid dynamics (CFD) analysis, resulting in a 9.77% increase in structural yield strength and a 23.8% reduction in noxious emissions.
 - Collaborated with process engineers to assess manufacturing feasibility and cost implications of implementing the proposed design.
- June–Aug 2020 **Research Assistant**, Data for Impact, Lehigh University, Bethlehem, PA
- Investigated the mechanical behavior of von Willebrand factor under applied forces to better understand its role in bleeding disorders.
 - Performed computational simulations to study protein deformation and elongation under varying mechanical conditions.
 - Analyzed simulation results to identify and generalize relationships between molecular-scale mechanics and clinically relevant bleeding behavior.

Research Activities

Collaborative Experimental Campaigns at Major Fusion Facilities

- March 2026 **Advanced Density Regulation for ITER and Future Reactors by Simultaneous Gas Puffing and Pellet Injection**, KSTAR (Korea Superconducting Tokamak Advanced Research), Korea Institute of Fusion Energy, Daejeon, South Korea
- Implemented an adaptive control framework in C within the real-time Plasma Control System, satisfying a $\sim 50 \mu s$ cycle-time constraint while modulating control effort to compensate for actuator input delays of $\sim 50\text{--}200 ms$.
 - Synchronized discrete and continuous actuators within a unified control framework by designing an integral-flow logic block to infer pellet injection state, enabling coordinated closed-loop gas and pellet fueling.
 - Validated closed-loop density regulation across 8 experimental runs on KSTAR, achieving sub-5% steady-state regulation error in collaboration with the KFE team.
- March 2026 **Proximity Control and Safe Reinforcement Learning Algorithms for Plasma Regulation with Stability-Limit Avoidance**, KSTAR (Korea Superconducting Tokamak Advanced Research), Korea Institute of Fusion Energy, Daejeon, South Korea
- Implemented a real-time RL controller in C to enable safe closed-loop exploration of high-density plasma scenarios.
 - Distributed the RL policy and real-time control loop across two CPUs with inter-processor communication, coupling a $\sim 500 \mu s$ RL inference subroutine to a $\sim 50 \mu s$ control cycle to prevent control dropouts and system crashes.
 - Demonstrated zero safety violations across 12 closed-loop experimental runs while tracking a time-varying density target, validating safety-prioritized control.
- June 2025 **Advanced Density Control in DIII-D for ITER and Beyond**, DIII-D National Fusion Facility, General Atomics, San Diego, CA, USA
- Prepared control-oriented models, simulation studies, and experimental procedures to support deployment of a model-reference adaptive density controller on DIII-D.
 - Coded and validated the adaptive controller in C within the real-time Plasma Control System, satisfying a $\sim 50 \mu s$ cycle-time constraint across multi-actuator coordination logic and actuator delay compensation.
 - Evaluated closed-loop control performance across multiple experimental scenarios in collaboration with the GA team.

Teaching and Supervision

- 2021–Ongoing **Student Mentor**, Lehigh Plasma Control Group, Lehigh University, Bethlehem, PA
- Guided junior researchers in developing simulation frameworks integrating physics-based and neural network surrogate models for dynamic modeling and predictive analysis.
 - Mentored junior researchers in the design and implementation of model-based and data-driven control strategies.
 - Supported preparation and delivery of findings at peer-reviewed conferences.
- 2018–2021 **Peer Tutor**, Center for Career & Professional Development, Lehigh University, Bethlehem, PA
- Provided one-on-one and small-group tutoring in core mechanical engineering courses, supporting students with fundamental concepts and problem-solving techniques.
 - Assisted students by reviewing coursework, clarifying challenging topics, and guiding them through analytical and computational problems.
 - Developed study guides and shared effective study strategies to reinforce foundational engineering knowledge and improve academic performance.

Clubs

- 2018–2021 **Designer**, Lehigh University Baja SAE
- Designed and validated the rear hub assembly for a competition off-road vehicle, iterating on geometry in SolidWorks to meet structural requirements.
 - Verified design integrity through FEA analysis, confirming the component met load-bearing specifications prior to fabrication.
 - Contributed to vehicle testing across varied terrain, evaluating performance against competition requirements.

Publications: Journal Articles

- [1] H. Al Khawaldeh, A. Pajares, S. T. Paruchuri, V. Graber, T. Rafiq, E. Schuster, J.-W. Juhn, T. Pederson, D. Shiraki, F. Turco, “Experimental Demonstration of Reactor-Relevant Adaptive Density Control on the DIII-D Tokamak with Coordinated Gas and Pellet Fueling,” submitted to *Plasma Physics and Controlled Fusion*, 2026.
- [2] S. T. Paruchuri, I. Ward, N. Rist, V. Graber, H. Al Khawaldeh, Z. Wang, E. Schuster, A. Pajares, and J.-W. Juhn, “Density Regulation With Disruption Avoidance in Next-Generation Tokamaks Using a Safe Reinforcement Learning-Based Controller,” *Fusion Engineering and Design*, vol. 216, 2025.
- [3] J. W. Berkery, (H. Al Khawaldeh), et al. (Collaboration Paper), “NSTX-U Research Advancing the Physics of Spherical Tokamak,” *Nuclear Fusion*, vol. 64, 2024.
- [4] S. T. Paruchuri, V. Graber, H. Al Khawaldeh, E. Schuster, “Dynamic Actuator Allocation via Reinforcement Learning for Concurrent Plasma Control Objectives,” *IEEE Transactions on Plasma Science*, vol. 52, 2024.
- [5] H. Al Khawaldeh, B. Leard, S.T. Paruchuri, T. Rafiq and E. Schuster, “Model-Based Linear-Quadratic-Integral Controller for Simultaneous Regulation of the Current Profile and Normalized Beta in NSTX-U,” *Fusion Engineering and Design*, vol. 192, 2023.

Publications: Refereed Conference Abstracts

- [1] H. Al Khawaldeh, S. T. Paruchuri, V. Graber, T. Rafiq, E. Schuster, A. Pajares, J. W. Juhn, S. H. Park, "Demonstration of Real-Time Multi-Actuator Plasma Density Control in KSTAR via Adaptive Fueling," *34th Symposium on Fusion Technology (SOFT)*, Aix-en-Provence, France, September 21-25, 2026.
- [2] H. Al Khawaldeh, V. Graber, S. T. Paruchuri, T. Rafiq, E. Schuster, "Coordinated Gas and Pellet Fueling for Plasma Density in KSTAR Under Reactor-Relevant Conditions," *67th Division of Plasma Physics (DPP) Annual Meeting of the American Physical Society (APS)*, Long Beach, CA, USA, November 17-21, 2025.
- [3] A. Pajares, H. Al Khawaldeh, F. Turco, D. Shiraki, T. Pederson, D. Eldon, J-W. Juhn, V. Graber, S. T. Paruchuri, E. Schuster, "ITER-Relevant Density Control with Pellets in DIII-Ds ITER Baseline Scenario," *67th Division of Plasma Physics Annual (DPP) Meeting of the American Physical Society (APS)*, Long Beach, CA, USA, November 17-21, 2025.
- [4] S. T. Paruchuri, H. Al Khawaldeh, V. Graber, E. Schuster, "Nonlinear Adaptive Control of Deuterium-Tritium Density Ratio in ITER," *67th Division of Plasma Physics (DPP) Annual Meeting of the American Physical Society (APS)*, Long Beach, CA, USA, November 17-21, 2025.
- [5] Y. Tao, B. Leard, H. Al Khawaldeh, T. Rafiq, E. Schuster, "Predictive Transport Modeling of KSTAR Advanced Scenarios Using COTSIM," *67th Division of Plasma Physics (DPP) Annual Meeting of the American Physical Society (APS)*, Long Beach, CA, USA, November 17-21, 2025.
- [6] Z. Wang, S. T. Paruchuri, H. Al Khawaldeh, T. Rafiq, E. Schuster, "Pellet Injection-Based Model Predictive Control of the Density Profile in Tokamaks by Leveraging Deep Reinforcement Learning," *31st IEEE Symposium on Fusion Energy (SOFE)*, Cambridge, MA, USA, June 23-26, 2025.
- [7] H. Al Khawaldeh, S.T. Paruchuri, E. Schuster, A. Pajares, and J. W. Juhn, "Adaptive Control of Plasma Density in Tokamaks Using Coordinated Gas Puffing and Pellet Injection," *31st IEEE Symposium on Fusion Energy (SOFE)*, Cambridge, MA, USA, June 23- 26, 2025.
- [8] H. Al Khawaldeh, S.T. Paruchuri, V. Graber, T. Rafiq, E. Schuster, A. Pajares, and J. W. Juhn, "Advanced Density Regulation for ITER Emulated Scenarios on DIII-D via Adaptive Control Techniques," *66th Division of Plasma Physics Annual (DPP) Meeting of the American Physical Society (APS)*, Atlanta, GA, USA, October 7-11, 2024.
- [9] S. T. Paruchuri, H. Al Khawaldeh, V. Graber, Z. Wang, N. Rist, I. Ward, T. Rafiq, E. Schuster, A. Pajares, J.W. Juhn, "Assessing Model-based and Learning-based Strategies for Active Density Regulation in ITER," *66th Division of Plasma Physics (DPP) Annual Meeting of the American Physical Society (APS)*, Atlanta, GA, USA, October 7-11, 2024.
- [10] H. Al Khawaldeh, S.T. Paruchuri, T. Rafiq and E. Schuster, "Simultaneous Regulation of the Dimensionless Gain and Density by Real-time Estimation of Confinement Times," *33rd Symposium on Fusion Technology (SOFT)*, Dublin, Ireland, September 22-27, 2024.
- [11] S. T. Paruchuri, H. Al Khawaldeh, V. Graber, E. Schuster, A. Pajares, J-W. Juhn, "Safe Reinforcement Learning-based Controller for Disruption Free Regulation of Plasma Density in Next-generation Tokamaks," *33rd Symposium on Fusion Technology (SOFT)*, Dublin, Ireland, September 22-27, 2024.
- [12] E. Schuster, S. T. Paruchuri, H. Al Khawaldeh, V. Graber, T. Sing, A. Pajares, J-W. Juhn, S. Park, "Advanced Model-based Density Control on KSTAR as Testbed for ITER Operation," *3rd International Fusion and Plasma Conference (iFPC)*, Seoul, South Korea, June 24-28, 2024.
- [13] H. Al Khawaldeh, S.T. Paruchuri, T. Rafiq and E. Schuster, "Model-Based Control of the Dimensionless Gain in KSTAR by Leveraging Real-time Estimation of the Confinement Time," *65th Division of Plasma Physics (DPP) Annual Meeting of the American Physical Society (APS)*, Denver, CO, USA, October 30-November 3, 2023.

- [14] H. Al Khawaldeh, S.T. Paruchuri, Z. Wang, T. Rafiq, E. Schuster, “Simultaneous Optimal Regulation of Kinetic+Magnetic Scalar Plasma Properties for Robust Sustainment of Advanced Scenarios in NSTX-U,” *29th IAEA Fusion Energy Conference*, London, UK, October 16-21, 2023.
- [15] L. Yang, H. Al Khawaldeh, S.T. Paruchuri, X. Song, Z. Wang, E. Schuster, “Towards Density Profile Regulation via Pellet Injection in Tokamaks Following a Hybrid Model Predictive Control Approach,” *29th IAEA Fusion Energy Conference*, London, UK, October 16-21, 2023.
- [16] J. W. Berkery, (H. Al Khawaldeh), et al. (Collaboration Paper), “NSTX-U Research Advancing the Physics of Spherical Tokamaks,” *29th IAEA Fusion Energy Conference*, London, UK, October 16-21, 2023.
- [17] H. Al Khawaldeh, B. Leard, S.T. Paruchuri, T. Rafiq and E. Schuster, “Model-Based Optimal Control of Core Kinetic+Magnetic Profiles and Scalar Plasma Properties in NSTX-U,” *64th Division of Plasma Physics (DPP) Annual Meeting of the American Physical Society (APS)*, Spokane, WA, USA, October 17-21, 2022.
- [18] B. Leard, C. Clauser, H. Al Khawaldeh, T. Rafiq, E. Schuster, “Development of COT-SIM+NUBEAMNet One-Dimensional Predictive Simulation Capability for Scenario Optimization in NSTX-U,” *64th Division of Plasma Physics (DPP) Annual Meeting of the American Physical Society (APS)*, Spokane, WA, USA, October 17-21, 2022.
- [19] H. Al Khawaldeh, B. Leard, S.T. Paruchuri, T. Rafiq and E. Schuster, “Model-Based Linear-Quadratic-Integral Controller for Simultaneous Regulation of Current Profile and Normalized Beta in NSTX-U,” *32nd Symposium on Fusion Technology (SOFT)*, Dubrovnik, Croatia, September 18-23, 2022.
- [20] H. Al Khawaldeh, B. Leard, S.T. Paruchuri, T. Rafiq, E. Schuster, “Model-based LQI Controller for Simultaneous Regulation of the Current Profile and Normalized Beta in NSTX-U,” *11th ITER International School on ITER Plasma Scenarios and Control*, San Diego, CA, USA, July 25-29, 2022.
- [21] H. Al Khawaldeh, A. Pajares, T. Rafiq, E. Schuster, “COTSIM-based Optimization of Transport Diffusivity Model for Predictive Simulations of NSTX-U Discharges,” *63rd Division of Plasma Physics (DPP) Annual Meeting of the American Physical Society (APS)*, Pittsburgh, PA, USA, November 8-12, 2021.
- [22] Y. Manjetta, H. Al Khawaldeh, T. Rafiq, E. Schuster, “COTSIM-based Gain Optimizer for Feedback Control of Advanced Scenarios in NSTX-U,” *63rd Division of Plasma Physics (DPP) Annual Meeting of the American Physical Society (APS)*, Pittsburgh, PA, USA (Remote), November 8-12, 2021.